

# AN INDONESIAN NEWS SENTIMENT INDEX FOR FORECASTING CONSUMER PRICE INDEX RATE

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**Abstract**—Accurate inflation nowcasting is hampered by data latency and translation-based sentiment limitations. Official Consumer Price Index (CPI) data are released with a delay, while translation-first sentiment pipelines distort linguistic nuance and reduce signal reliability. This study constructs the Indonesia News Sentiment of Inflation (INSEI) using a native-language student model (IndoLEM) distilled from Gemini 2.5 Flash. Forecasting results for 2024-2025 show that news sentiment operates primarily as a leading indicator. Concurrent models failed to explain inflation variance, while lagged Vector Autoregression achieved the lowest error with an RMSE of 0.49, outperforming both autoregressive and teacher-based benchmarks. While linear estimators underestimated the March 2025 inflation spike of 1.65% by approximately one percentage point, the results validate native-language sentiment as a scalable framework for inflation analysis in emerging markets.

**Index Terms**—Inflation Forecasting, News Sentiment Analysis, Indonesian NLP, Knowledge Distillation

## I. INTRODUCTION

Accurate and timely data are essential for effective inflation management. In Indonesia, price stability is the primary objective of monetary policy under the inflation targeting framework adopted by Bank Indonesia [1]. Inflation is officially measured by the monthly Consumer Price Index (CPI) published by Badan Pusat Statistik (BPS) [2]. However, CPI figures are released with a monthly delay [3]. Consequently, the ability of monetary authorities to respond to rapidly changing price dynamics is constrained. Delays in the official CPI releases motivate inflation nowcasting using alternative high-frequency data, including indicators derived from news text [4][5].

To mitigate these limitations, researchers and policymakers have increasingly turned to high frequency big

data from online sources, such as news media and article summaries [6]. Transforming large volumes of unstructured text into a quantitative sentiment index offers a promising opportunity for inflation nowcasting. Despite this potential, sentiment analysis research in the Southeast Asian context has been hindered by a critical methodological flaw, namely the reliance on translating native language text into English before applying sentiment analysis tools [7]. In particular, a lexicon-based study on inflation in Vietnam explicitly noted that translation altered sentiment polarity and reduced scoring reliability [7]. Similarly, research on Indonesian inflation sentiment using Twitter data found that translating Indonesian text into English before using the VADER lexicon could distort semantic meaning and sentiment interpretation [8]. Such translation-first pipelines risk losing linguistic nuance, cultural context, and idiomatic expressions, thereby undermining the validity of sentiment based indices.

Beyond translation-related limitations, existing studies exhibit a significant conceptual gap in their treatment of inflation as a single, monolithic concept [9]. Most recent methodologies construct general sentiment indices using broad keywords such as “inflation”, “price increase”, or “CPI” implicitly assuming that public sentiment about inflation is uniform [6]. In practice, however, public concern and media discourse are often driven by sharp price movements in specific, highly visible categories of goods and services. For the average Indonesian household, aggregate CPI movements are abstract, whereas sudden increases in fuel prices (BBM), basic foodstuffs (harga pangan), or electricity tariffs (tarif listrik) represent tangible and immediate economic pressures that dominate the news coverage and public discussion.

This study addresses these gaps by constructing the Indonesia News Sentiment of Inflation (INSEI), a native-language sentiment index by knowledge distillation. Unlike translation-based or computationally expensive

proprietary approaches, we distill the reasoning capabilities of a large teacher model (Gemini 2.5 Flash) into a compact, open-source student model optimized for the Indonesian economic text. To the best of our knowledge, this work represents the first application of teacher-student distillation for Indonesian inflation forecasting that explicitly avoids translation-based sentiment modelling.

## II. LITERATURE REVIEW

The use of artificial intelligence for inflation forecasting has expanded rapidly with advances in machine learning and natural language processing, enabling textual data to supplement traditional macroeconomic indicators. Early studies using social media sentiment, including post-COVID-19 inflation discussions on Twitter analyzed with methods such as K-Nearest Neighbors and Support Vector Machines, demonstrate that public sentiment reflects inflation-related concerns and can serve as an early warning signal for shifts in price expectations [8]. Similar findings are reported in Romania, where incorporating sentiment indicators improved inflation forecast accuracy, and post-pandemic evidence suggests that changes in inflation discourse may track CPI dynamics more closely than conventional indicators [9][10].

More recent literature emphasizes the role of language model based sentiment measures as forward looking signals in short-term inflation forecasting. Sentiment indices derived from economic news have been shown to outperform traditional macroeconomic variables during periods of heightened uncertainty, reflecting expectation-driven price dynamics. Evidence from Vietnam demonstrates that a lexicon-based sentiment index constructed from economic news can predict CPI movements, confirming that economic sentiment acts as a leading indicator of inflation and suggesting cross-country applicability of text-based approaches [7].

As economic data increasingly originate from text-rich environments, researches have explored how quantitative signals complement numerical time-series models[11]. Studies of post-COVID-19 inflation sentiment highlight that public concern about inflation often shifts sharply during economic stress, providing early signals ahead of CPI releases [7]. Survey of research on social-media and news analytics shows that machine-learning models applied to textual data uncover behavioral patterns and expectations that are not visible in purely numerical datasets [12][13].

In addition, machine-learning based extractive summarization of high-frequency news has proven effective in capturing economic shocks more rapidly than traditional data releases, offering gains when combined with classical forecasting models [14]. Broader financial studies demonstrate that news sentiment influences investor behavior and market volatility, reinforcing the predictive value of text-based indicators beyond macroeconomic forecasting [15]. Applied to the Indonesian context, sentiment analysis of the inflation-related discussions confirms that public sentiment can be quantified and correlated with CPI movements, supporting the integration of text-derived indicators into inflation forecasting frameworks [16].

Finally, hybrid and data-filtered sentiment analysis approaches further enhance forecasting robustness by isolating economically relevant signals, reducing noise inherent in raw text data [17]. Evidence from UK newspaper sentiment shows that news-based indicators contain forward-looking information on inflation and broader macroeconomic outcomes, including growth and uncertainty, often leading conventional variables [18]. Collectively, these studies underscore the growing importance of high-frequency textual sources such as news articles, policy communications, and online commentary in modern inflation forecasting systems.

Recent advances in knowledge distillation validate teacher-student learning as a robust methodology for efficient natural language processing, where large language models generate labeled datasets to supervise compact student models. Performance-guided and targeted distillation frameworks show that LLM-generated labels enable smaller models to preserve, and in some cases exceed, teacher-level task performance while achieving substantial reductions in inference cost latency [19][20].

## III. RESEARCH METHODOLOGY

### A. Overview

This paper presents a new sentiment-based economic indicator for Indonesia that merges news article classification with time-series forecasting. The methodology encompasses four stages: data acquisition and filtering, text preprocessing and label generation, multi-model training and alignment validation, and index construction and forecasting. Three transformer-based models (IndoBERT-base-p2, IndoLEM/IndoBERT-base-uncased, mDeBERTa-v3-base) are trained using distillation from Gemini 2.5 Flash and evaluated against monthly CPI inflation rates covering January 2017 to October 2025.

### B. Data Collection and Filtering

The ground truth time-series for inflation analysis is the Month-to-Month (M-to-M) Consumer Price Index (CPI) rate for Indonesia. This data, encompassing the period from January 2017 to December 2025, was sourced from Badan Pusat Statistik (BPS). The dataset contains the monthly percentage change in the CPI, which is utilized for subsequent correlation and time-series modeling with the extracted news data.

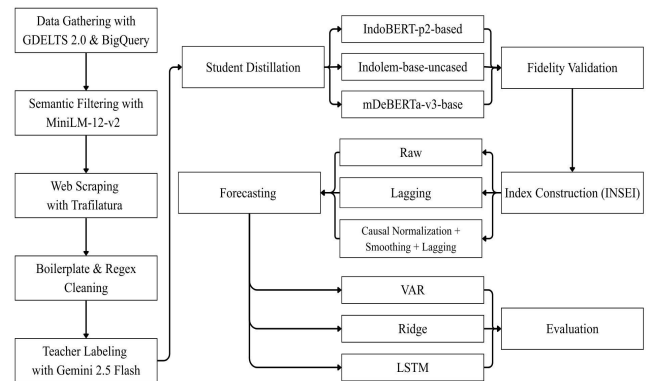


Fig. 1. Research Methodology Flowchart

News articles are extracted from the GDELTS 2.0 GKG dataset via BigQuery using a query targeting five Indonesian news domains (bisnis.com, liputan6.com, cnnindonesia.com, tempo.co, detik.com) published between 2017 and 2025. The query filters for economic themes and geographic mentions of Indonesia, yielding 380,167 URLs.

### C. Preprocessing

URLs and headlines were preprocessed and semantic filtering was performed using the SentenceTransformer paraphrase-multilingual-MiniLM-L12-v2 model. Manually curated inflation headlines were used as reference vectors, and articles with cosine similarity scores of at least 0.7 were retained. Article bodies and metadata were then retrieved through a custom web scraper with *Trafilatura* based scraping pipeline. To ensure the linguistic quality required, a rigorous preprocessing pipeline was applied to the 9,226 articles extracted. First, the text underwent Unicode normalization (NFKC) to standardize characters with standardization of dash characters and the reduction of excessive whitespace. Then, a domain-specific cleaning by identity and boilerplate removal. Articles resulting in fewer than 50 characters post-cleaning were discarded to ensure sufficient information density for semantic analysis. Furthermore, to optimize computational efficiency and adhere to input length constraints of the subsequent embedding models, the text content was truncated to a maximum of 3,000 characters. The resulting curated corpus contained 8,992 articles.

### D. Labelling

The creation of the gold standard corpus required the semantic labeling of each article with respect to its dominant economic sentiment regarding Indonesian inflation. This was done automatically using the Gemini 2.5 Flash model through a custom Python application using batch Application Programming Interface (API) calls. This model was set to work under the persona of a macroeconomic analyst with expertise in Indonesian Consumer Price Index. The input for the model was further standardized into the first 1,000 characters of the cleaned article text in an effort to reduce computation latency and cost and at the same time retain the core semantic meaning. The classification task was initiated by the following prompt in Indonesian as well as its translation in English for documentation:

*“Kamu adalah analis ekonomi makro yang menilai berita tentang inflasi di Indonesia khususnya pada index Indeks Harga Konsumen (CPI) di Indonesia. Untuk setiap teks berita berikut, berikan: Label: Inflation, Deflation, atau Neutral. Alasan singkat (1 kalimat) mengapa kamu memilih label itu.”*

*“You are a macroeconomic analyst evaluating news related to inflation in Indonesia, specifically the Consumer Price Index (CPI). For each news text, provide: Label: Inflation, Deflation, or Neutral. A brief one-sentence rationale justifying your assigned label.”*

The resulting corpus contained 8,992 articles, distributed as 4,443 Neutral, 3,221 Inflation, and 1,328 Deflation. A manual review of all the labeled samples was performed to verify classification accuracy.

### E. Model Training and Alignment

To construct the domain classifiers, three distinct pre-trained transformer architectures were selected for fine-tuning: IndoBERT-base-p2 as the baseline, IndoLEM/IndoBERT-base-uncased represents domain specific Indonesian corpora, and mDeBERTa-v3-base represents strong multilingual contextualization. These models function as “student” networks, fine-tuned to approximate the “teacher” (Gemini 2.5 Flash) classification logic. The dataset was partitioned into a stratified split of 70% training (6,294), 15% validation (1,349), 15% test (1,349). Training was implemented using the HuggingFace Transformers library with a sequence truncation length of 512 tokens and dynamic padding. Given the uneven distribution of economic sentiments, a weighted cross-entropy loss function with label smoothing ( $\alpha = 0.1$ ) was implemented during the training phase.

Hyperparameters were empirically selected under strict memory constraints and limitations on available computational power, thus precluding the execution of exhaustive grid search with the following configuration: batch size 8, gradient accumulation steps 4, gradient checkpointing enabled, learning rate  $2e-5$ , cosine learning rate scheduling with 10% warmup, weight decay 0.01, maximum 10 epochs with early stopping (patience 3), with FP16 mixed precision.

### F. Fidelity and Alignment Validation

A separate holdout dataset of 1,908 articles (collected but unseen during training) is labeled by Gemini 2.5 Flash and evaluated by all three student models. Fidelity metrics include accuracy, macro-F1, Cohen's kappa, and Jensen-Shannon divergence between predicted probability vectors and teacher labels. High teacher student alignment ensures that downstream sentiment indices preserve the semantic decision boundaries of the teacher model.

### G. Index Construction

The construction of the Indonesia News Sentiment of Inflation (INSEI) differs for the Teacher and Student models, reflecting their respective outputs. Neutral sentiment is excluded as it conveys informational absence rather than directional price pressure. Since the Gemini 2.5 Flash model provided discrete classifications the teacher index  $INSEI_{T,t}$  is constructed based on the net proportion of articles in each category for a given month  $t$ . Where  $N$  represents the count of articles assigned to each respective label. This results in a step-wise index that reflects the voting consensus of the teacher model.

$$INSEI_{T,t} = \frac{N_{Inflation,t} - N_{Deflation,t}}{N_{Total,t}} \quad (1)$$

$$INSEI_{S,t} = \frac{1}{M_t} \sum_{i=1}^{M_t} (P(Inf)_i - P(Def)) \quad (2)$$

The student models output a probability distribution over the  $P_{Neut}$ ,  $P_{Inf}$ ,  $P_{Def}$  classes. To capture subtle shifts in economic sentiment confidence, the student index  $INSEI_{S,t}$  is defined as the mean difference in class probabilities. Where  $M_t$  is the number of articles in month  $t$ . By utilizing soft

labels, the student index preserves the intensity of the sentiment, providing a smoother and more granular signal than the discrete teacher index. To evaluate predictive utility, three index variations were constructed. First, the Raw Index ( $INSEI_{Raw}$ ) represents the unmodified sentiment signal derived directly from net probabilities scores. Second, the Raw-Lagged Index ( $INSEI_{Lagged}$ ) incorporates a predictive horizon of  $k = 2$  months, selected via Granger Causality testing ( $P < 0.05$ ). Third, the Final Processed Index ( $INSEI_{Final}$ ) ensures stationarity through a sequential pipeline: Expanding Window Z-score Normalization (window = 6), 3-month moving average smoothing, a  $k = 2$  month lag shift, and rescaling to a baseline of 50 following standard diffusion conventions.

$$INSEI_{final,t} = 50 + (10 \times INSEI_{norm,t-2}) \quad (3)$$

#### H. Forecasting and Evaluation

Three distinct forecasting architectures were implemented. Autoregressive (AR) model served as the baseline, while a Vector Autoregression (VAR) model was employed for the hybrid bivariate specification [Rate, INSEI], with the optimal lag structure determined dynamically by the Bayesian Information Criterion (BIC). Ridge Regression was used to address multicollinearity in 12-month lagged inflation and sentiment with regularization optimized with 5 fold cross-validation. To capture non-linear dependencies, a Long Short-Term Memory (LSTM) network was designed. The architecture consists of a 64 units layer and a dropout rate of 0.1, followed by a fully connected dense layer of 16 units. Training was conducted for a maximum of 50 epochs using a 12-month lookback window, utilizing an early stopping mechanism.

All models were trained on 78 samples covering January 2017 to December 2023 and evaluated on 22 out-of-sample observations from January 2024 to October 2025.

### IV. RESULTS AND ANALYSIS

#### A. Sentiment Classification Performance

Table I presents the classification metrics. The validation subset serves as a structural sanity check and is not intended to reflect task performance. The observed low accuracy parallels random probability of 33%, while the low F1-Macro confirms the model’s initially unrefined state. These near-random baselines explicitly necessitate the subsequent fine-tuning phase to align the semantic representations with the target classes. Post fine-tuning, IndoLEM achieves the optimal F1-Macro (0.74) and exhibits balanced effectiveness with the highest Recall (0.76) and Precision (0.73). These metrics indicate robust class discrimination and minimized false positives, and shows that domain specific pre-training on formal Indonesian corpora (IndoLEM) provides superior semantic capture compared to multilingual pre-training for this specific task.

Performance consistency varied across sentiment classes. Even though all models demonstrated high precision and recall on the Neutral class, the minority Deflation class (14.8% of samples) presented a significant challenge, with precision scores ranging from (0.52 to 0.60) and recall scores ranging from (0.68 to 0.71). IndoLEM achieves the highest

TABLE I. MODEL CLASSIFICATION PERFORMANCE

| Model                | F1 Macro    | Recall Macro | Precision Macro |
|----------------------|-------------|--------------|-----------------|
| Pre-trained Baseline |             |              |                 |
| IndoBERT-base-p2     | 0.18        | 0.33         | 0.20            |
| IndoLEM              | 0.23        | 0.32         | 0.32            |
| mDeBERTa-v3-base     | 0.22        | 0.33         | 0.16            |
| Fine-Tuned           |             |              |                 |
| IndoBERT-base-p2     | 0.70        | 0.72         | 0.69            |
| IndoLEM              | <b>0.74</b> | <b>0.76</b>  | <b>0.73</b>     |
| mDeBERTa-v3-base     | 0.73        | 0.74         | 0.72            |

TABLE II. MODEL CONFIDENCE SCORE

| Model            | Neutral     | Inflation   | Deflation   |
|------------------|-------------|-------------|-------------|
| Indobert-base-p2 | 0.60        | 0.57        | 0.97        |
| IndoLEM          | 0.84        | 0.82        | 0.98        |
| mDeBERTa-v3-base | <b>0.64</b> | <b>0.91</b> | <b>0.96</b> |

Deflation class recall (0.71), indicating greater sensitivity to rare deflation signals but struggles on precision (0.59). The difficulty is amplified by the relatively small proportion of deflation samples in the training corpus.

Table II shows confidence evaluation identifies IndoLEM as the most robustly calibrated model, maintaining high probability scores across all samples. IndoLEM offers a superior balance of accuracy and predictive certainty across all the tests.

#### B. Teacher-Student Fidelity

Table III shows the behavioral alignment between the student models and the teacher (Gemini 2.5 Flash) was evaluated on a hold-out new set of 1,908 articles. Model mDeBERTa-v3-base achieved the strongest fidelity, recording the highest Cohen’s Kappa and the lowest Jensen-Shannon Divergence, indicating the closest approximation to the teacher’s probability distribution. However, higher teacher-student alignment does not necessarily imply superior performance, as the effectiveness depends on how the sentiment signal interacts with the time series model. Teacher-student correlation across all NLP models exceeded 0.97, confirming that architectural differences resulted in minimal semantic variation when trained on identical teacher labels. Moreover, the temporal alignment observed in Fig. 2 shows minimal inter-model variance across sentiment index trajectories, demonstrating strong architectural consistency despite fluctuations in the target inflation rate.

TABLE III. TEACHER-STUDENT ALIGNMENT METRICS

| Model            | Accuracy    | Kappa       | Js Divergence |
|------------------|-------------|-------------|---------------|
| Indobert-base-p2 | 0.80        | 0.47        | 0.34          |
| IndoLEM          | 0.81        | 0.48        | 0.33          |
| mDeBERTa-v3-base | <b>0.81</b> | <b>0.51</b> | <b>0.33</b>   |

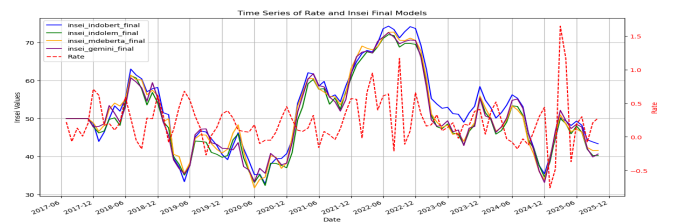


Fig. 2. Temporal alignment of ( $INSEI_{Final}$ ) student model predictions against the official CPI Inflation Rate.

### C. Sentiment Index Calibration

Table IV shows the correlation that raw state based indices exhibited moderate linear correlations with the CPI rate. However, the significant reduction in correlation to the range of 0.17 to 0.20 following Expanding Window Z-score Normalization indicates that the initial strength was largely driven by common non-stationary trends rather than immediate structural co-movement. This transformation was applied to enforce approximate stationarity, thereby preventing spurious regression results. Although the contemporary correlation is lower, Fig 3 shows that Granger Causality tests confirmed that the stationary signal possesses significant predictive power at Lag-2 ( $P < 0.05$ ) across all models. This result validates that the predictive relationship at Lag-2 is statistically robust.

The 12-month rolling correlation illustrated in Fig. 4 demonstrates that the INSEI captures critical structural breaks in inflation dynamics. A sharp negative divergence (-0.75) characterizes late 2019, preceding the pandemic. In contrast, a rapid structural realignment, related to the onset of COVID-19 in 2020, started a steep surge in correlation that reached 0.50 by 2021. This upward momentum strengthened during subsequent shocks, with distinct peaks during the 2022 fuel price adjustments and the 2023-2024 El-Niño events, confirming the index's responsiveness to volatility.

TABLE IV. INSEI CORRELATION WITH CPI INFLATION RATE

| Correlation with CPI | IndoBERT | IndoLEM | mDeBERTa | Gemini |
|----------------------|----------|---------|----------|--------|
| $INSEI_{Raw}$        | 0.31     | 0.33    | 0.35     | 0.32   |
| $INSEI_{Lagged}$     | 0.17     | 0.18    | 0.18     | 0.16   |
| $INSEI_{Final}$      | 0.17     | 0.20    | 0.19     | 0.20   |

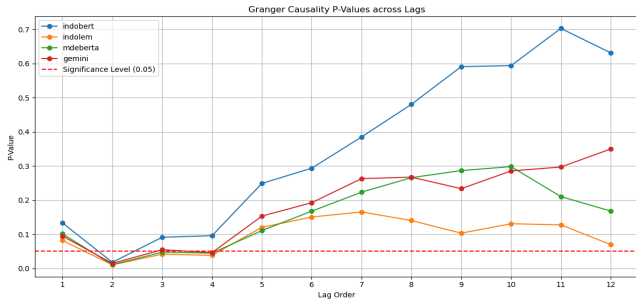


Fig. 3. Granger causality p-values across lag structures. The threshold at  $k = 2$  confirms that the INSEI sentiment signals statistically predicts CPI inflation at 2-month horizon ( $P < 0.05$ ) across all architectures.

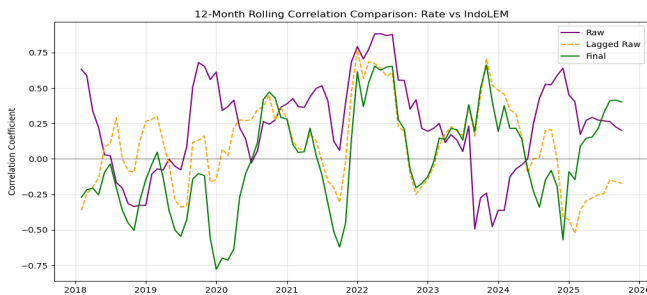


Fig. 4. Rolling Correlation dynamics showing pre-pandemic divergence, a sharp realignment during COVID-19, and volatility-driven peaks during 2022-2024 supply shocks.

### D. Inflation Forecasting Results

Performance was evaluated using out-of-sample forecasts over a 22-month period from January 2024 to October 2025. Forecast accuracy is assessed primarily through error-based metrics rather than explained variance, which is typically low in short-horizon inflation forecasting. Preliminary analysis showed that raw indices produce negative  $R^2$  values, while lagged specifications consistently minimized RMSE.

For this reason, Table V reports the lagged configuration, to isolate the index's leading indicator properties. The most robust performance was obtained by the Vector Autoregression (VAR) framework using the IndoLEM-derived index, achieving an RMSE of 0.49 and a modest  $R^2$  of 0.02 outperforming both the baseline and teacher-based (Gemini) model. These findings suggest domain-specialized student models can effectively distill macroeconomic signals without generative computational overhead. In contrast, Ridge Regression and LSTM architectures failed to generalize, which is likely driven by the limited effective sample size relative to model complexity. The forecast trajectory on Fig. 5 illustrates that while the VAR-IndoLEM model accurately captures the mean directional shifts and the late 2024 deflationary cycle, extreme volatility, particularly the 1.65% inflation spike in March 2025 is substantially underestimated by the linear estimator. This confirms INSEI as a leading indicator, while linear models remain conservative in high-variance regimes.

TABLE V. FORECASTING PERFORMANCE LAGGED INSEI

| Configuration      | MAE         | RMSE        | R Squared    |
|--------------------|-------------|-------------|--------------|
| <b>VAR</b>         |             |             |              |
| Baseline(AR Only)  | 0.35        | 0.50        | 0.00         |
| Hybrid (Gemini)    | 0.36        | 0.50        | 0.01         |
| Hybrid (IndoBERT)  | 0.36        | 0.50        | 0.00         |
| Hybrid (IndoLEM)   | <b>0.36</b> | <b>0.49</b> | <b>0.02</b>  |
| Hybrid (mDeBERTa)  | 0.36        | 0.50        | 0.01         |
| <b>LSTM</b>        |             |             |              |
| Baseline (AR Only) | <b>0.35</b> | <b>0.50</b> | <b>0.00</b>  |
| Hybrid (Gemini)    | 0.36        | 0.50        | 0.00         |
| Hybrid (IndoBERT)  | 0.36        | 0.50        | -0.01        |
| Hybrid (IndoLEM)   | 0.36        | 0.50        | 0.00         |
| Hybrid (mDeBERTa)  | 0.36        | 0.50        | -0.01        |
| <b>Ridge</b>       |             |             |              |
| Baseline (AR Only) | 0.62        | 0.75        | -0.16        |
| Hybrid (Gemini)    | 0.61        | 0.74        | -0.14        |
| Hybrid (IndoBERT)  | 0.61        | 0.74        | -0.14        |
| Hybrid (IndoLEM)   | <b>0.58</b> | <b>0.71</b> | <b>-0.04</b> |
| Hybrid (mDeBERTa)  | 0.59        | 0.73        | -0.10        |

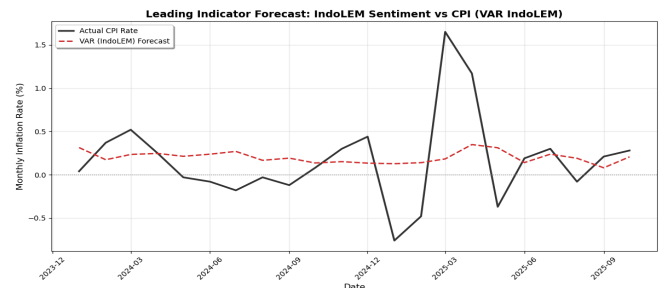


Fig. 5. VAR-IndoLEM forecast versus CPI Rate, capturing direction but underestimating extreme volatility.

## V. CONCLUSIONS

This study addresses both the methodological limitations of translation-based sentiment analysis and the conceptual weakness of treating inflation as a monolithic construct. We introduce a native-language news sentiment index built on a gold-standard corpus of 9,226 Indonesian articles, demonstrating that knowledge distillation can effectively bridge resource constraints in developing economies. Comparative evaluation shows that the Indolem-based student model achieves the strongest forecasting performance within a VAR framework, delivering the lowest RMSE of 0.49 without sacrificing predictive fidelity relative to the teacher model.

Crucially, the results isolate the temporal role of economic sentiment. News sentiment does not explain inflation variance, but lagged specifications consistently improve forecast accuracy, confirming sentiment as a leading indicator rather than a concurrent explanatory variable. By processing full-text semantics in the native language, the proposed index captures heterogeneous price pressures that are obscured by translation-based and keyword-driven approaches.

The system demonstrates strong directional accuracy and timely identification of the late-2024 deflationary cycle. However, linear VAR models underestimate extreme inflation shocks, most notably the March 2025 spike, indicating that while native-language sentiment robustly captures trend dynamics, linear frameworks remain conservative under high volatility.

Future work should extend the historical baseline, explore non-linear hybrid architectures to better capture tail risk, and decompose the aggregate sentiment index into sector-specific sub-indices. Such extensions would enable finer isolation of inflation drivers while preserving the computational efficiency of the student model.

## AUTHOR'S CONTRIBUTIONS

A.T.: Conceptualization, Methodology, Formal Analysis, Investigation, Visualization, Writing. J.H.W.: Conceptualization, Data Curation, Investigation, Writing. N.T.M.S.: Supervision. H.L.: Supervision. R.C.P.: Supervision..

## AVAILABLE DATA AND MATERIALS

The complete source code, including the data gathering pipeline and model notebooks, is available in the <https://github.com/arieltarliman/AN-INDONESIAN-NEWS-SENTIMENT-INDEX-FOR-PREDICT-CPI-RM->. The official Consumer Price Index Rate (CPI) data utilized in this study is publicly accessible through the Central Bureau Statistics (BPS) Indonesia at <https://www.bps.go.id/id/statistics-table/2/MSMy/inflasi-bulanan-m-to-m-.html>

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